

EE326 Intro. to Information Theory and Coding

Final, Spring 2018

Date: Monday, Jun. 11, 2018

Time: 9am–12pm

Problem 1 (15 points)

A Z-channel is depicted in Fig. 1 below for $\epsilon \in [0, 1]$. Let the input distribution be denoted by $P_X(0) = 1 - p$, $P_X(1) = p$ for $p \in [0, 1]$.

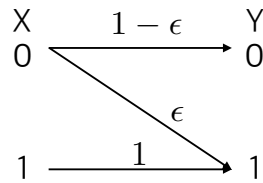


Figure 1: Z-channel

- (5 points) Write an expression for the mutual information $I(X; Y)$ as a function of p and ϵ . Draw a few sketches of the mutual information, one as a function of ϵ for a fixed p and the other as a function of p for a fixed ϵ . In your sketches, pay attention to the extreme points, convexity, and illustrate whether the symmetric input distribution $P_X = [1/2, 1/2]$ achieves the capacity or not. (You don't have to solve the mutual information maximization analytically.)
- (3 points) Next consider a cascaded version of two Z channels, as shown in Fig. 2. Here the output of the first channel is directly fed into the second channel (without re-encoding). Find the transition probabilities $P_{Z|X}$ of the overall channel $X \rightarrow Z$.

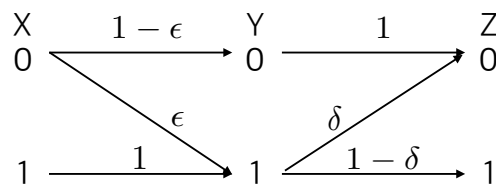


Figure 2: Cascaded Z-channels

- (7 points) Denote the capacity of the first Z-channel by $C_{X \rightarrow Y}$, the capacity of the second channel by $C_{Y \rightarrow Z}$, and the capacity of the cascaded channel in Fig. 2 by $C_{X \rightarrow Z}$. Remind that the capacity is defined as the maximum mutual information of a given channel, maximized by all possible input distributions. Show that

$$C_{X \rightarrow Z} \leq \min\{C_{X \rightarrow Y}, C_{Y \rightarrow Z}\}.$$

Find conditions for ϵ and δ where the equality holds.

Problem 2 (25 points)

In class, we discussed Forney's derivation of error exponent for a BSC(ϵ), with random coding. In this problem, we study an idea of improving the error exponent by modifying the random codes.

We will use a BSC(ϵ) with $\epsilon < 1/2$. Consider (n, R) codes, composed of 2^{nR} codewords, denoted by $X^n(1), X^n(2), \dots, X^n(2^{nR})$. Let these codewords be independently generated by i.i.d. symbols of Bernoulli($1/2$) distribution. Suppose that $X^n(1)$ is transmitted by n -uses of a BSC(ϵ) with $\epsilon < 1/2$. For the channel output Y^n , we compute the probability of error with a ML decoder, which finds $\arg \min_{m \in \{1, \dots, 2^{nR}\}} |Y^n \oplus X^n(m)|$ where $|Y^n \oplus X^n(m)|$ is the Hamming weight (the number of 1's) of $Y^n \oplus X^n(m)$.

- a) (4 points) Let $N(\gamma)$ denote the expected number of codewords $X^n(j)$, $j \neq 1$, which have a Hamming distance $n\gamma$ from $X^n(1)$, i.e., $|X^n(j) \oplus X^n(1)| = n\gamma$. Show $N(\gamma)$ can be written as

$$N(\gamma) \doteq 2^{nR} 2^{-nD_B(c_1 \| c_2)} \quad (1)$$

where $D_B(\alpha \| \beta) = \alpha \log \frac{\alpha}{\beta} + (1 - \alpha) \log \frac{1 - \alpha}{1 - \beta}$. **Find c_1 and c_2 .**

- b) (4 points) We next consider the pairwise error event between $X^n(1)$ and a particular incorrect codeword $X^n(j)$, which is $n\gamma$ -distance away from $X^n(1)$. Show that the probability that the ML decoder prefers $X^n(j)$ to $X^n(1)$ is given in the form of

$$\Pr(\hat{X}^n = X^n(j) | X^n(1)) \doteq 2^{-n\gamma D_B(c_3 \| c_4)}. \quad (2)$$

Find c_3 and c_4 .

Hint: ML prefers $X^n(j)$ to $X^n(1)$ iff at least a half of $n\gamma$ bits where $X^n(j)$ differ from $X^n(1)$ are flipped by the BSC(ϵ).

- c) (4 points) Show that the average error probability of the random codes with ML decoder is upper bounded by

$$P_e^{(n)} \leq 2^{-nE_1(R)} \quad (3)$$

where

$$E_1(R) = \min_{\gamma \in [0, 1]} [D_B(c_1 \| c_2) - R + \gamma D_B(c_3 \| c_4)], \quad (4)$$

for c_1, c_2 in (1) and c_3, c_4 in (2).

- d) (6 points) Denote the solution of γ in (4) as γ^* (You don't need to calculate this value). Now consider a new encoding: a modified random coding with an expurgation process. After we generate 2^{nR} codewords by random coding, we compare every pair of codewords and eliminate one of the codewords if the Hamming distance between the two codewords is less than $n\gamma_{GV}(R)$. By doing this, we guarantee $N(\gamma) = 0$ for $\gamma < \gamma_{GV}(R)$, i.e., every pair of codewords have at least $n\gamma_{GV}(R)$ distance from each other. (Remind that $\gamma_{GV}(R)$ is the Gilbert-Varshamov distance satisfying $D_B(\gamma_{GV}(R) \| 1/2) = R$.) Under what condition on R , will such an expurgation process strictly improve the error exponent in (4)? Write the condition on R in terms of γ^* . Explain the reasonings of your answer to get the full score.
- e) (7 points) The above expurgation process can be viewed as a way to guarantee that all codeword pairs have at least $n\gamma_{GV}(R)$ distance. Explain why it does not make sense to do the expurgation process for an even larger $\gamma > \gamma_{GV}(R)$, which may guarantee an even larger distance between every pair of codewords.